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PALEOCENE VIRTUAL GEOMAGNETIC POLES DETERMINED FROM VOLCANIC ROCKS NEAR GOLDEN, COLORADO

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Three flat-lying basalt flows near Golden, Colorado dated at about 58 m.y. by K-Ar methods have a stable thermoremanent magnetization. The virtual geomagnetic pole position after AC cleaning is located at N 81.4°, E 63.7° for the upper two flows and N 64.5°, E 62.9° for the lowest flow. These poles are comparatively close to previously reported early Tertiary pole positions from North America (exclusive of the Siletz River Volcanics pole).

Magnetic stability of the samples is directed related to the degree of high-temperature oxidation of opaque minerals.

1. Introduction

Relatively few early Tertiary paleomagnetic poles have been determined from radiometrically dated North American rocks (see Irving [1], appendix). In this note we report two virtual geomagnetic poles determined from samples of basaltic volcanic rocks near Golden, Colorado one flow of which Everden et al. [2] dated at 58.7 m.y. by K-Ar analysis.

2. Geologic relations

The samples we have studied were collected from North Table Mountain, just east of the Colorado Front Range, at Golden, Colorado. Here, three essentially flat-lying lava flows are intercalated with continental sedimentary rocks above the paleontologically defined Cretaceous-Tertiary boundary in the Denver Formation (Van Horn [3]). The stratigraphic relations of the flows are shown diagrammatically in fig. 1. The volcanic rocks are porphyritic trachybasalts or mafic latites composed of phenocrysts of augite and andesine-labradorite from 1 to 5 mm in length in a holocrystalline very-fine grained groundmass of plagioclase, K-feldspar, augite, olivine, apatite, biotite, and opaque iron-titanium oxide minerals. Chlorite, serpentine, clay, calcite, zeolites, hematite, and quartz occur as deuteric and supergene alteration products.

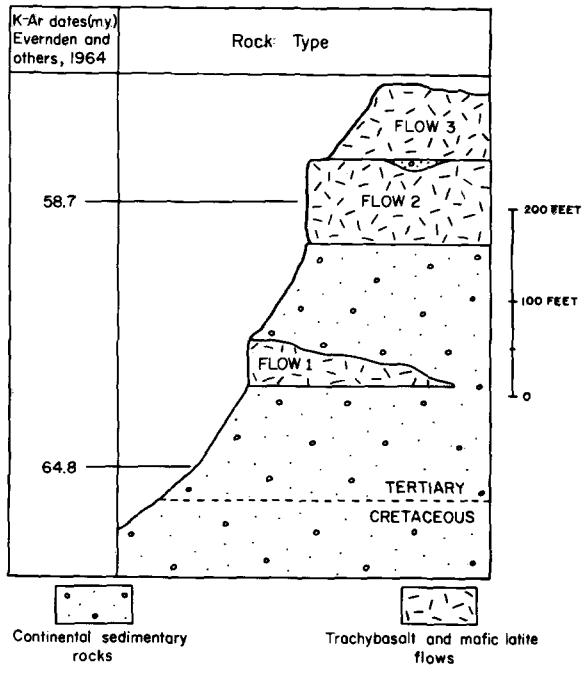


Fig. 1. Generalized stratigraphic section of upper part of Denver Formation near Golden, Colorado.

Evernden et al. [2] obtained K-Ar dates from the Denver Formation on South Table Mountain, about two miles southeast of where we collected samples for

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paleomagnetic study. They report an age of 64.8 m.y. for plagioclase from a bed of dacitic pumice fragments which contains Puercan mammals and Denver flora about 35 feet stratigraphically above the Cretaceous-Tertiary boundary, an an age of 58.7 m.y. for whole rock from a basaltic flow about 235 feet stratigraphically above the dated pumice bed. The dated flow on South Table Mountain can be readily correlated with the North Table Mountain section where it is equivalent with our flow 2 (fig. 1).

3. Techniques

Bulk samples oriented with a magnetic compass were collected from outcrops of flows 1, 2 and 3 on North Table Mountain. In the laboratory one-inch cores were drilled from the samples and NRM directions were measured on a astatic magnetometer with a sensitivity of about 10^{-5} cgs/cm^3. The cores were subsequently cleaned by partial AC demagnetization in fields up to 250 Oe to remove soft components of the NRM, and magnetic directions and intensities were measured on a spinner magnetometer with a sensitivity of 10^{-7} cgs/cm^3. Susceptibilities were determined with a Soiltest model MS-3 susceptibility bridge.

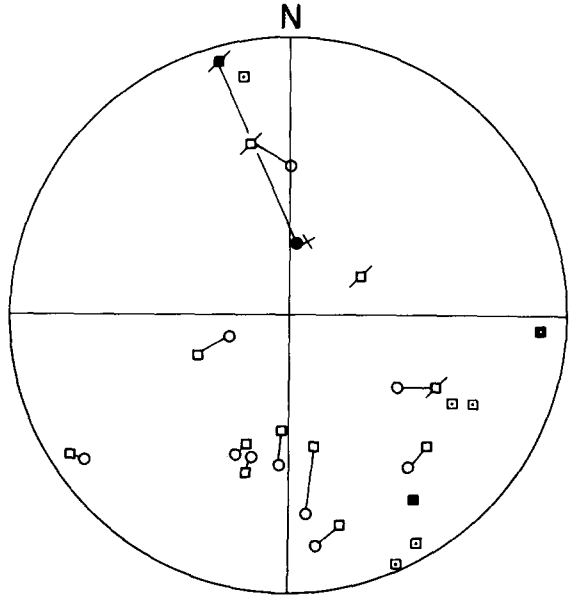


Fig. 2. NRM directions and results of storage test. Squares are NRM directions, circles are directions after storage. Lines connect directions from same sample. Samples from flow 1 marked with a dot in center of symbol. Samples omitted from statistical analysis marked with ticks. Open symbols for points on upper hemisphere, solid symbols for points on lower hemisphere. Cross indicates present inclined dipole.

4. Paleomagnetic results

NRM directions are shown in fig. 2. The samples are predominantly reversely magnetized, but display only a slight tendency toward grouping. Storage of the samples in random orientations in the earth's field for 18 months produced some changes in magnetic directions (see fig. 2). Susceptibilities ranged from 1.4 to 18.0 $\times$ 10^{-5} cgs/cm^3. Generally, those samples showing the greatest movement during the storage test have susceptibilities greater than 6×10^{-5} cgs/cm^3.

All samples were subjected to AC cleaning in peak fields of 250 Oe. As shown in fig. 3, the grouping of directions was greatly improved by partial demagnetization.

Four samples with magnetic vectors significantly away from the grouping were comparatively strongly magnetized and AC cleaning in peak fields of 450 Oe effected little movement from their initial positions. It is concluded that these samples have been remagnetized by lightning. Inasmuch as two of the flows we sampled cap the topographic high of North Table Mountain and summer thunderstorms are common in the region, this seems a reasonable conclusion. Because of their widely divergent nature these four samples were omitted from statistical consideration.

Fisher statistics have been applied to the fifteen grouped samples and a summary of the results is given in table 1.

There is a definite correlation between magnetic stability and degree of oxidation of opaque minerals. Polished section examination at 1000X magnification reveals that the principal magnetic minerals formed by primary magmatic precipitation were titanomagnetite and ilmenite occurring either as discrete grains, or intergrown. Titanomagnetite grains

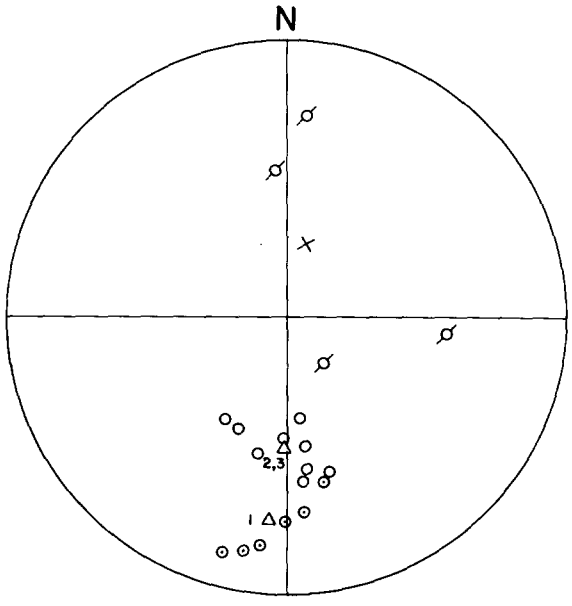


Fig. 3. Directions of magnetization after partial demagnetization in 250 Oe field. Open circles are directions on upper hemisphere. Samples from flow 1 marked with a dot in center of circle. Samples omitted from statistical analysis marked with ticks. Triangles are mean directions. Mean for flow 1 (1); mean for flows 2 and 3 (2,3). Cross indicates present inclined dipole.

up to 250μ are present, but crystals less than 5μ are abundant. Post-crystallization high-temperature oxidation has variably affected the flows. The primary opaques of flow 3 are generally highly oxidized; ilmenite lamellae, metailmenite, hematite, and pseudobrookite are very common phases. Samples from this flow possess low susceptibilities and small components of unstable magnetization. In contrast, flow 2 is variably oxidized. Some samples show little or no effect of high-temperature oxidation; others display moderate effects including ilmenite lamellae and rarely metailmenite. The samples showing little high-temperature oxidation are commonly affected in varying degrees by low-temperature oxidation, that is, titanomaghemitization. Samples largely unaffected by high-temperature oxidation generally possess comparatively high susceptibilities and large components of unstable magnetization.

These observations are in accord with the ideas of Larson et al. [4] who show that increased stability of magnetization is related to grain-size effects and degree of high-temperature oxidation.

5. Discussion

The computed virtual geomagnetic poles are shown in fig. 4. We have followed the suggested usage of Cox and Doell [5] in defining these as virtual rather than paleo geomagnetic poles.

The remanent directions from flows 2 and 3 are closely grouped and probably indicate that these two flows erupted within a short time span about 58.7 m.y. ago. One virtual pole was calculated from these two flows. The magnetic directions from flow 1 have

Table 1
Paleomagnetic data, North Table Mountain after 250 Oe demagnetization.

Flow	No. of specimens	Site latitude	Site longitude	Mean D	Mean I	Precision factor k	$\alpha 95$
2 and 3	9	N . 40°47′	W . 105°13′	182	−50	46.9	7.6°
1	6	N . 40°47′	W . 105°13′	185	−27	39.3	10.8°

Virtual magnetic pole

Flow	Paleolatitude	Paleolongitude	δM	δP
2 and 3	N . 81.4°	E . 63.7°	6.9	10.2
1	N . 64.5°	E . 62.9°	6.4	11.8

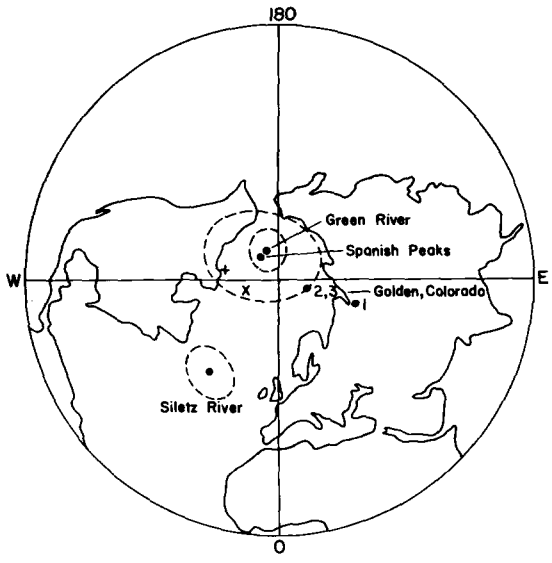


Fig. 4. Early Tertiary magnetic pole positions, North America. Dashed lines outline 95 percent confidence limits for authenticated poles. Virtual geomagnetic pole for Golden flow 1 (1); virtual geomagnetic pole for Golden flows 2 and 3 (2,3). Plus sign is the present pole position; cross is the pole position determined from the present magnetic field in central Colorado.

much lower inclinations than the top two flows and a separate virtual geomagnetic pole was calculated for this flow. The flatter inclinations of remanent directions from flow 1 as compared with flows 2 and 3, the presence of about 100 feet of clastic sedimentary rock between flows 1 and 2, and the position of flow 1 80 to 90 feet above a tuff bed 6 m.y. older than flow 2 (see fig. 1) suggest that a considerable length of time elapsed between the eruption of flow 1 and flow 2. Unfortunately the continental clastic sedimentary rocks separating the flows are dominantly conglomerates from which we could not obtain stable magnetic directions, or siltstones from which we were unable to collect coherent samples.

Authenticated poles from rocks of early Tertiary age in North America are also shown on fig. 4. The Spanish Peaks pole and the Green River Formation pole are relatively close to the poles we have determined, and all are relatively close to the present north rotational pole. In contrast, the pole from the Siletz River Volcanics [8] lies far from these and is considered unusual. Possibly the Siletz River Volcanics have been tectonically rotated since eruption; or perhaps during cooling of the Siletz flows the main dipole field made a transient excusion away from the rotational axis.

Acknowledgements

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References

[1] E.Irving, Paleomagnetism and its application to geological and geophysical problems (Wiley, New York, 1964) p. 399.
[2] J.F.Everden, D.E.Savage, G.H.Curtis and G.T.James, Potassium-argon dates and the Cenozoic mammalian chronology of North America, Am. Jour. Science 262 (1964) 145.
[3] R.Van Horn, Bedrock geology of the Golden quadrangle, Colorado, U.S. Geol. Survey map GQ-103 (1957).
[4] E.E.Larson, Ozima, Mituko, Ozima, Minoru, T.Nagata and D.W.Strangway, Stability of remanent magnetization of rocks, Geophys. J. 17 (1969) 262.
[5] A.Cox and R.R.Doell, Review of paleomagnetism, Geol. Soc. Am. Bull. 71 (1960) 645.
[6] E.E.Larson and D.W.Strangway, Magnetization of the Spanish Peaks dike swarm, Colorado and Shiprock dike, New Mexico, J. Geophys. Res. 74 (1969) 1505.
[7] O.W.Torreson, T.Murphy and J.W.Graham, Magnetic polarization of sedimentary rocks and the earth's magnetic history, J. Geophys. Res. 54 (1949) 111.
[8] A.Cox, Remanent magnetism of Lower to Middle Eocene basalt flows from Oregon, Nature 179 (1957) 685.